

Hardik Gupta

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EDUCATION

PES University , <i>B.Tech in Electronic and Communication Engineering</i>	Aug 2023 – Present — Bangalore, India
Lotus Valley International School , <i>Class 12th- CBSE</i>	Apr 2022 – Mar 2023 — Noida, India
Lotus Valley International School , <i>Class 10th- CBSE</i>	Apr 2020 – Mar 2021 — Noida, India

CERTIFICATION

Foundations of Data Structures & Algorithms in Python – Packt (Coursera) Jul 2026
– Mastered core **data structures**, **algorithmic efficiency**, and spatial/temporal complexity optimization in Python.

AI & ML with Drone Technology - IIT Hyderabad (Tihan) Jun 2025 – Feb 2026
– Hands-on training in **ML**, **Deep Learning**, and AI deployment with academic exposure from **IIT Hyderabad**.
– Built and evaluated **regression**, **classification**, **clustering**, and **neural network** models using Python libraries.
– Gained practical understanding of the building blocks and working principles of drones, including core components.

Microsoft Office Specialist: Excel 2013 – Microsoft
– Certified in spreadsheet data management, complex formulas, pivot tables, and data modeling in MS Excel.

EXPERIENCE

Bharat Space Education Research Centre, New Delhi Dec 2025 – Jan 2026
Winter Intern
– Modeled aircraft stability and performance using **MATLAB (AID toolbox)** and **XFLR5**, identifying key design parameters for optimal flight behavior.

ColorHome Lighting, Noida Jul 2025 – Aug 2025
Summer Intern
– Analyzed the end-to-end manufacturing lifecycle of 9W LED bulbs, from component sourcing to final dispatch.
– Gained hands-on experience in **SMT operations**, **quality assurance**, and **electrical safety testing**.

PROJECTS

Vocab Ledger – Native Android Vocabulary App ▶ Jul 2026
– Architected an offline-first native Android app using **Kotlin**, **Jetpack Compose**, **Room DB**, and **Firestore**, featuring B-Tree database indexing and cloud sync; published in Closed Testing on **Google Play Store**.
– Optimized release binary size by **44%** (from 19.7 MB down to 11.0 MB) by configuring custom **R8/ProGuard** minification and resource shrinking rules.

Automated PCB Defect Detection System 🤖 Apr 2026
– Trained and benchmarked **YOLOv5s** and **YOLOv8s** on 693 PCB images to detect and classify 6 manufacturing defect types, achieving **97.4% mAP@0.5** and **95.2% recall**.
– Built a 4-level severity analysis system that evaluates defect area, position, and confidence to deliver automated per-board PASS/FAIL verdicts with repair recommendations.

Automatic Audio Synthesis for Silent Videos 🎧 Mar 2026
– Implemented the AutoFoley Frame-Relation Network (**ResNet-50 + Multiscale TRN**) to predict and synthesize audio for silent videos across 10, 12, and 15 sound-class configurations using VGGSound and AVSync15 datasets.
– Built an end-to-end pipeline covering frame extraction, per-class average spectrogram computation, temporal relation-based classification, and Griffin-Lim audio reconstruction from predicted spectrograms.

Federated AI for Privacy-Preserving Alzheimer's Detection 🧠 Aug 2025 – Present
– **Designing a secure Federated Learning pipeline** for Alzheimer's MRI classification using **VGG19/ResNet50**, enabling decentralized model training on private data.
– **Integrating Grad-CAM and SHAP** to generate visual heatmaps and feature importance scores, improving the interpretability of the model's predictions.

SKILLS

Languages & Concepts: Python, C, Kotlin, Assembly, Data Structures & Algorithms (DSA), Jetpack Compose, Room DB, Firebase
Machine Learning & CV: Scikit-learn, TensorFlow, PyTorch, OpenCV, YOLOv5/v8, NumPy, Pandas, Matplotlib, Seaborn
Tools & Frameworks: Git, Jupyter, VS Code, MATLAB, Google Colab, Power BI, **MS Excel**, Ripes, Arduino IDE, LaTeX